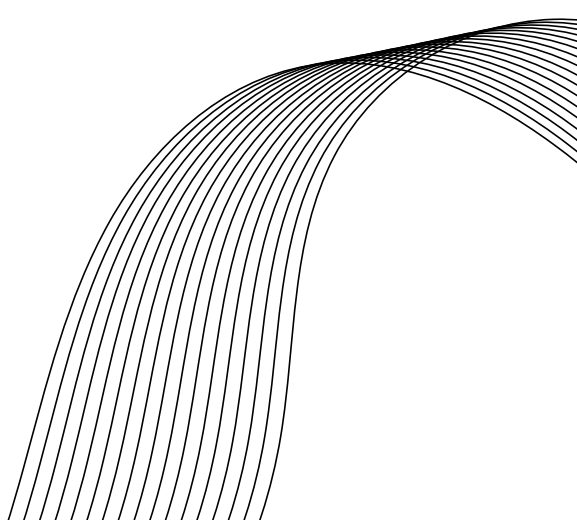
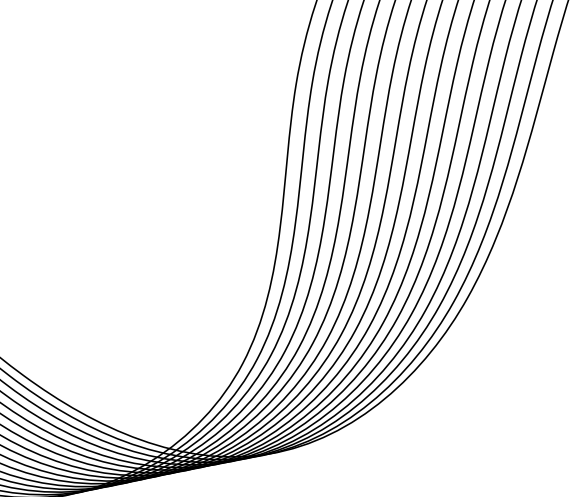




TEAM IMPROMPTU INTELLIGENCE (I.I.)



LAND IS A FINITE RESOURCE

- We can't manufacture more land.
- Over **33%** of the world's soils are already degraded due to **unsustainable** farming practices

If we lose our soil, we lose our ability to feed the future.





WHY THAT MATTERS



- Without action, the UN warns food production could drop by 40% by 2045
- Movements like Save Soil are uniting billions to protect this living resource
- This isn't just a climate issue, it's a food security, livelihood, and humanity issue.

Healthy soil = healthy food = a healthy, sustainable future.

THE FARMER STRUGGLE

Behind every field is a family, behind every harvest is hope.



✓ **No more young farmers**
Over 67% of Nova Scotian farmers are 55 or older.

✓ **The Manual" Way**
Manually juggling multiple sources on spreadsheets and paper.

✓ **Farms are decreasing**
In Nova Scotia, the number of farms has dropped by 21% since 2016

✓ **Limited Access to Knowledge**
Limited or no access to modern, sustainable, and data-driven farming tools

WHAT'S AVAILABLE RIGHT NOW



Large-Scale Enterprise Solutions

Powerful, but expensive and
hardware-driven

Open Scientific Models

Complex for the average
farmer to use.

The “Manual” Way

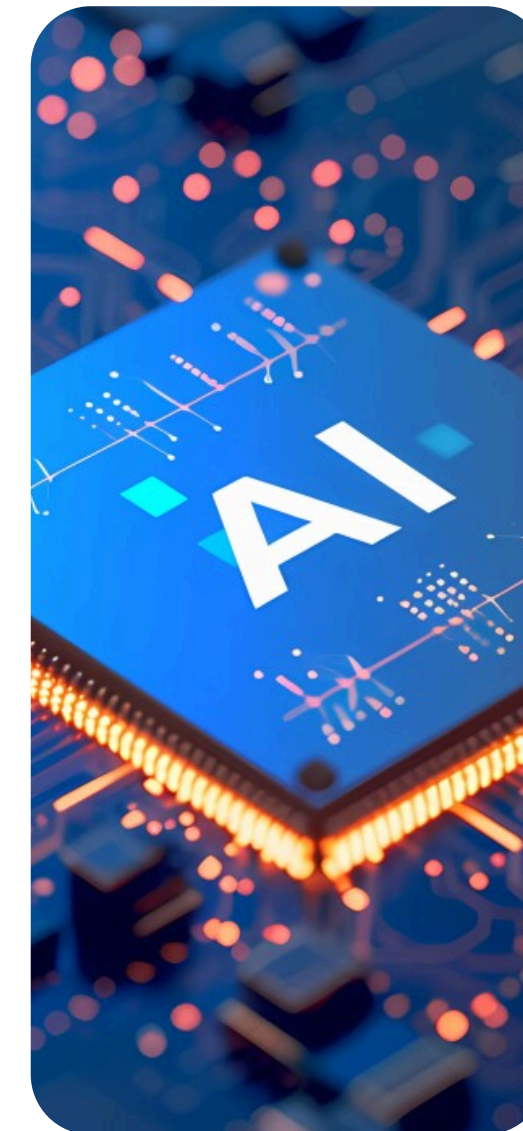
Manually juggling multiple
sources on spreadsheets and
paper.



SOILSENSE: SUSTAINABLE FARMING INTELLIGENCE

1. **Upload** a photo of your **soil** and enter **land size**
2. **Local** K-Nearest Neighbours model **identifies** soil **type**
3. App recommends **best crops** based on **soil, location,** and **season.**
4. Provides **insights** on **cost efficiency** and **land-use optimization.**

Healthy soil = healthy food = a healthy, sustainable future.



TURNING SOIL INTO SMART DECISIONS

Impact

- Empower beginner farmers with accessible, data-driven guidance
- Promotes sustainable land use and reduces soil degradation

Innovation

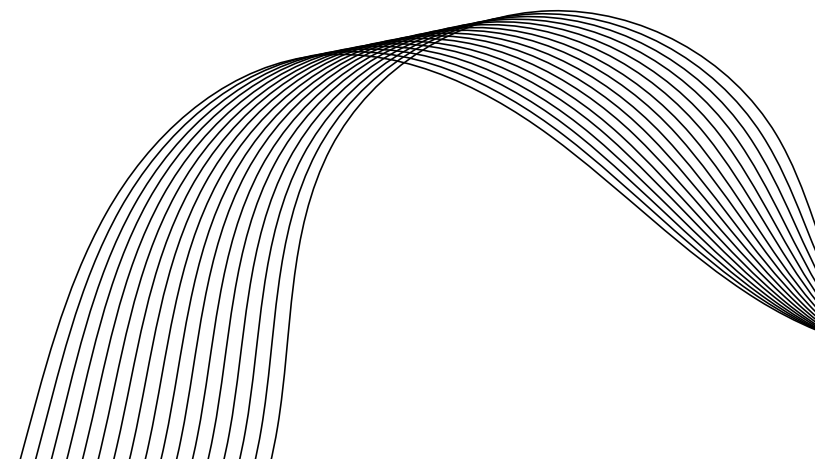
- Combines AI soil analysis, geolocation data, and seasonal insights
- Designed for low-resource or rural areas with lightweight models

Creativity

- Turns soil images into personalized farming recommendations
- Future potential:
 - Model Improvements
 - Plant health tracker



A DEMO OF SoilSense



THANK YOU!

Team I.I. (Impromptu Intelligence):

- Parth Bindra
- Rabjot Kaur
- Sami Yakan
- Paras Nath Seth

